Learning Objectives

- To learn about the self-propelled vehicle with stage by stage improvement from past to present.

4.0 INTRODUCTION

4.1 HISTORY OF AUTOMOBILE

In the early days, the man began his journey by walking. Then he used animals like horse, elephant, camel, donkey for his journey. Wheels are the most ancient discovery for human kind. With the help of wheel, the mankind designed horse carriage, bullock cart for transportation of people and goods. As time has gone on, they have devised increasingly more effective and efficient methods of travel. The automobile made a dramatic change in the way people travel.

Leonardo da Vinci considered the idea of a self-propelled vehicle in the 15thcentury. In 1680, Sir Issac Newton discovered that if steam is sent out in the rear direction, the vehicle will move forward. In 1769, Nicolas-Joseph Cugnot of France was the constructor of the first true automobile. Cugnot's vehicle was a steam-powered tri cycle carrying four people, and run for 20 minutes at3.6 km per hour.

During the 18th century, James Watt invented the steam engine and it leads to many developments in road transportation. In 1801, Richard Trevithick of Great Britain invented a steam-powered road carriage. Following him, W.H. James invented the automobile running with different speed. Till then, the research was focused on the External combustion engine. In 1863, Jean-Joseph-Etienne Lenoir, a Belgian engineer invented the "horseless carriage" and it uses an internal combustion engine. This is the first commercially successful internal combustion engine. In 1867, Nikolaus August Otto, German engineer invented the four-stroke internal combustion engine. This engine is the first to efficiently burn fuel directly in a piston chamber.

Two-stroke internal combustion engine was invented by Sirclerk, a German scientist in 1880. In 1885, a German engineer, Karl Benz builds the first true automobile powered by a gasoline engine. It has three wheels and looked similar to a carriage. In 1886, Gottlieb Wilhelm Daimler and Wilhelm Maybach invent the first four-wheeled, four-stroke engine in Germany. It is known as the "Cannstatt-Daimler."

In 1892, Rudolf Diesel was a German thermal engineer, invented the internal-combustion engine that runs on diesel. In 1894, Benherd developed an automobile by placing the engine in the front part of the chassis.

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